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EDUCATION

- Korea Advanced Institute of Science and Technology (KAIST)**
Ph.D. Candidate. CCS Graduate School of Mobility
M.S. CCS Graduate School of Mobility
Advisor: Prof. Namwoo Kang, (**Homepage:** smartdesignlab.org)
Daejeon, Korea
Mar. 2023 - Feb. 2027
Aug. 2020 - Feb. 2023
- Korea University of Technology and Education**
B.S. Mechatronics Engineering
Cheonan, Korea
Feb. 2012 - Aug. 2019

EXPERIENCE

- Graduate Researcher**
Smart Design Lab., KAIST
Daejeon, Korea
Sep. 2020 - Present
- AI Researcher**
Narnia Labs
Daejeon, Korea
Feb. 2023 - Aug. 2024
- Mechanical Design Engineer**
ATI | Advanced Technology Inc.
Incheon, Korea
May. 2019 - Oct. 2019
- Student Researcher**
Computer-Aided-Engineering Lab., KOREATECH
Cheonan, Korea
Apr. 2017 - Dec. 2018

INDUSTRIAL PROJECTS

- Samsung Heavy Industries**
Generative AI-Based Sloshing Load Prediction and CCS Configuration Optimization
Apr. 2026 - Oct. 2026
- Hyundai Motor Group**
Development of a Battery Failure Prediction Algorithm Based on Anomaly Detection
Development of a GNN-Based Deep Learning Library for Collision Analysis Prediction
GNN-Based Deep Learning for Performance Prediction from 3D Geometry Data
Generative Design Guidelines and AI-Based Classification for Automotive Road Wheels
Feb. 2024 - Mar. 2025
Sep. 2023 - Dec. 2023
Jul. 2022 - Jun. 2023
Jul. 2021 - Jul. 2022
- LG Display**
Optimization of Panel Layered Structures Using Generative Design and MeshGraphNet
May. 2022 - Feb. 2023
- Korea Expressway Corporation**
Development of an Expressway Accident Risk Index for Identifying Hotspots
Mar. 2021 - Dec. 2021
- Korea Railroad Corporation (KORAIL)**
Transportation Performance Analysis and Modal Substitutability under COVID-19
Sep. 2020 - Feb. 2021

PUBLICATIONS

- Park, J.**, & Kang, N.* (2026). "AutoATC: Automated Analytical Target Cascading Decomposition with LLM-Based Multi-Agent System." (Under review)
- Park, J.**, & Kang, N.* (2026). "LLM-Guided Adaptive Penalty Parameter Updates for Analytical Target Cascading." (Under review)
- Park, J.**, & Kang, N.* (2026). "Point-DeepONet: Predicting Nonlinear Fields on Non-Parametric Geometries under Variable Load Conditions." *Neural Networks*, 108560. ([PDF](#)) ([code](#)) ([dataset](#))
- Kim, J., **Park, J.**, Kim, N., Yu, Y., Chang, K., Woo, C. S., Yang, S.*, & Kang, N.* (2025). "Physics-Constrained Graph Neural Networks for Spatio-Temporal Prediction of Drop Impact on OLED Display Panels." *Expert Systems with Applications*, 126907. ([PDF](#))
- Hong, S., Kwon, Y., Shin, D., **Park, J.**, & Kang, N.* (2025). "DeepJEB: 3D Deep Learning-Based Synthetic Jet Engine Bracket Dataset." *Journal of Mechanical Design*, 147(4). ([PDF](#)) ([dataset](#))
- Park, J.**, & Kang, N. (2024). "BMO-GNN: Bayesian Mesh Optimization for Graph Neural Networks to Enhance Engineering Performance Prediction." *Journal of Computational Design and Engineering*, 11(6), 260-271. ([PDF](#))
- Yun, J., Lee, J., **Park, J.**, Chung, K., & Lee, J.* (2022). "How to Measure the Network Vulnerability of Cities to Wildfires: Cases in California, USA." *Transportation research record*, 2676(12), 382-395. ([PDF](#))

INTERNATIONAL CONFERENCES

- **Park, J.**, & Kang, N.* (2026). “LLM-Guided Adaptive Penalty Parameter Updates for Analytical Target Cascading”, In *Asian Congress of Structural and Multidisciplinary Optimization (ACSMO 2026)*.
- **Park, J.**, & Kang, N.* (2025). “A PointNet-Enhanced Operator Network for Nonlinear Analysis of Non-Parametric 3D Geometries Under Varying Load Conditions”, In *16th World Congress of Structural and Multidisciplinary Optimisation (WCSMO 16)*.
- Kim, J., **Park, J.**, Yang, S., & Kang, N.* (2024). “Design Optimization of Multi-layered Display Panels for Drop Impact using Physics-Constrained Graph Neural Network”, *The 26th International Congress of Theoretical and Applied Mechanics 2024 (ICTAM2024)*.
- **Park, J.**, & Kang, N.* (2023). “Bayesian Mesh Optimization for Graph Neural Networks to Enhance Engineering Performance Prediction”, In *ASME 2023 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC/CIE 2023)*.
- **Park, J.**, & Kang, N.* (2023). “How to Apply AI to Real-World Design Problems”, In *ASME 2023 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC/CIE 2023)*.
- Yun, J., Lee, J.Y., **Park, J.**, Chung, K., and Lee, J.* (2022), “How to Measure the Vulnerability of Cities to Wildfires: Cases in California”, *Transportation Research Board (TRB) 101st Annual Meeting*.

DOMESTIC CONFERENCES

- Cho, Y., **Park, J.**, & Kang, N.* (2026). “A Data-Driven Framework for Railway Track Condition Assessment Using the Analytic Hierarchy Process”, In *The Korean Society of Mechanical Engineers (KSME)*.
- **Park, J.**, & Kang, N.* (2025). “DialogCAD: Interactive CAD Generation Framework with a Multi-Agent System via Model Context Protocol”, In *The Korean Society of Mechanical Engineers (KSME)*.
- **Park, J.**, Kim, J., Jang, W., Lee, Y., Lee, S., & Kang, N.* (2025). “A Study on 3D Topology Optimization Shape Reconstruction with CSG-Based Deep Learning”, In *The Korean Society of Mechanical Engineers (KSME)*.
- **Park, J.**, & Kang, N.* (2025). “SG-DeepONet: A Spatial Gradient-Aware Deep Operator Network for Pressure and Shear Stress Prediction in Vehicle Aerodynamic Analysis”, In *The Korean Society of Mechanical Engineers (KSME)*.
- **Park, J.**, & Kang, N.* (2024). “Stress Field Prediction in Nonlinear Static Analysis for Arbitrary 3D Geometries Using Latent-DeepONet”, In *The Korean Society of Mechanical Engineers (KSME)*.
- Hong, S., Kwon, Y., Shin, D., **Park, J.**, and Kang, N.* (2024). “DeepJEB: 3D Deep Learning-based Synthetic Jet Engine Bracket Dataset”, In *The Korean Society of Mechanical Engineers (KSME)*.
- Shin, D., **Park, J.**, Seo, M., & Kang, N.* (2024). “3D Field Prediction and Uncertainty Quantification with Implicit Neural Representation”, In *The Korean Society of Mechanical Engineers (KSME)*, 289-290.
- **Park, J.**, Kim, J., Yu, Y., Kim, N., Ha, S., Jang, K., & Kang, N.* (2023). “A Study on Impact Analysis based on Graph Neural Networks for Optimization of the Display Panel Structure”, In *The Korean Society of Mechanical Engineers (KSME)*, 144-145.
- **Park, J.**, & Kang, N.* (2022). “Bayesian Mesh Optimization for Graph Neural Networks”, In *The Korean Society of Mechanical Engineers (KSME)*, 1626-1628.
- **Park, J.**, Shin, D., Yoo, S., & Kang, N.* (2022). “Preliminary Study on 3D Wheel Performance Prediction Using Graph Neural Networks”, In *The Korean Society of Mechanical Engineers (KSME)*, 316-317. (Best Paper)
- Yun, J., Lee, J., **Park, J.**, Chung, K., & Lee, J.* (2021). “How to Measure the Vulnerability of Cities to Wildfires, Cases in California”, In *Korean Society of Transportation*, 436-437.
- Mok, H., **Park, J.**, Lee, J., Han, D., & Kim, D.* (2021). “Development of Express Accident Risk Index (ex-ARI) for Identifying Hotspots”, In *The Korean Society of Civil Engineers (KSCE)*, 57-58.
- Mok, H., Lee, J., Han, D., **Park, J.**, & Kim, D.* (2021). “Establishment of Traffic Safety Measures based on Accident Risk Index(ex-ARI), Cases in Dangjin Yeongdeok Expressway”, In *The the Korean Society of Civil Engineers (KSCE)*, 506-507.
- **Park, J.**, Kim, D., Mok, H., Han, D., & Lee, J.* (2021). “Identifying Fatal and Severe Accident Hotspots based on Multiple Safety Performance Measures”, In *The Korean Society of Civil Engineers (KSCE)*, 49-50. (Best Paper)

AWARDS

- | | |
|---|----------|
| • Best Presentation Award , Korean Society of Mechanical Engineers | May 2022 |
| • Best Presentation Award , Korean Society of Civil Engineering | Nov 2021 |
| • Best Poster Award , Korean Society of Civil Engineering | Nov 2021 |
| • Minister of Science, Technology & Telecommunication Award , Capstone Design Fair | Oct 2018 |

- **Korea Institute for Robot Industry Advancement Award**, SEOULTECH Robot Competition Oct 2018
- **Chancellor's Award, Kyungnam University**, Changwon National Robot Competition Sep 2018
- **KOREATECH Industrial-Academic Cooperation Award**, SolidWorks Technology Competition Dec 2017
- **Chancellor's Award, KOREATECH**, University Engineering Design Contest Nov 2017

OTHERS

- **TA**, KAIST MO833 Lecture: "AI-based Mobility Design", Overview of DeepONet Applications Nov. 2024
- **TA**, KAIST MO833 Lecture: "AI-based Mobility Design" Mar. 2024 - Jun. 2024
- **TA**, KAIST Industry-Academic Collaboration Lecture: "Analysis Prediction to Optimization with AI" Feb 2024
- **Patent**, KR 10-2025-0091992 May 2025
Apparatus and Method for Interactive CAD Generation Based on a Multi-Agent System via a Model Context Protocol
- **Patent**, Wheelchair Maneuverability Enhancement, KR 10-2018-0015617 Feb 2018